CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EE)

April/May 2012

EE 202 ELCTRICAL POWER GENERATION

Date: 02.05.2012, Wednesday	Time: 01:30 p.m. To 04:30 p.m.	Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) What are the factors of selection of site for hydroelectric station? [07]

How the hydroelectric plants are classified?

Q - 2 (a) Explain with neat schematic diagram working of nuclear power plant. [14]

(b) What are the factors of selection of site for nuclear power plant? Explain advantages and disadvantages of a nuclear power plant.

OR

Q - 2 (a) Explain the classification and components of gas turbine power plant. [14]

(b) Explain steam water cycle of thermal power plant with neat sketch.

Q - 3 (a) Explain wind turbine generator unit with battery storage facilities. [5]

(b) Give the main criteria in selection of site for wind farm. [2]

(c) Explain with well labeled diagram components of diesel engine power plant. [7]

OR

Q - 3 (a) Draw the sketch of MHD generator and describe each component in detail. [14]

(b) Explain any one fuel cell with its advantages and disadvantages.

SECTION – II

Q - 4 Explain the different components of steam power plant with its function. Also give the advantages and disadvantages of steam power plant. **[07]**

Q - 5 (a) Explain the origin of geothermal resources. **[3]**

(b) Give the classification of geothermal electric power plants according to the geothermal fluid used and thermodynamic cycle adopted. **[4]**

(c) With well labeled diagram explain the binary cycle liquid dominated geothermal power plant. **[7]**

OR

Q - 5 (a) Explain how with neat sketch the use of tidal power is made to generate electricity. **[06]**

(b) Explain fuel cycle of thermal power plant. **[03]**

(c) Explain the selection of site for steam power plant. **[05]**

Q - 6 (a) With well labeled diagram explain the working of a closed cycle OTEC plant. **[14]**

(b) Give the advantages and limitations of ocean energy conversion.

OR

Q - 6 (a) With well labeled diagram explain binary cycle solar thermal power plant. **[14]**

(b) Explain working principle of solar pond with neat sketch.

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